

xviii) The only device that operates at all layers of the OSI model is _

A) Network host

C) Router

B) Switch

D) Bridge

xix) Which network topology involves a use of a central hub for connections?

A) Mesh

C) Tree

B) Bus

D) Star

xx) While browsing the internet, you open up 4 different tabs on chrome browser. In the first tab, you are on Facebook.com. On the second tab, you are streaming live music on a certain website. In the third tab, you open twitter.com. In the fourth tab, you have musbsep.mubs.ac.ug open. Why is that information from the Facebook tab never mix up with information in mubsep.mubs.ac.ug tab yet you are using the same web browser and accessing different web sites and servers.

A) The operating system comes to our rescue and opens up logical connection bearing a unique session ID for each tab opened

B) Your web browser is intelligent enough to handle the tasks without any issue

C) The transport layer by design ensures that such a task is accomplished that way

D) The session layer creates physical ports for every connection.

xxi) The application layer of the TCP/IP model is responsible for creating, maintaining and terminating sessions on the network

A) TRUE: _____

B) FALSE: _____

xxii) Not all users are people but any device that requires a given service from another network component can be a user

A) TRUE: _____

B) FALSE: _____

xxiii) The only network services provided by campus networks are DHCP, NAT, DNS, Web Server, Mail Server, File and Print Server

A) TRUE: _____

B) FALSE: _____

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xiii) When IP addresses exist in the numeric format of AAA.BBB.CCC.DDD. Each address consists of four _____ separated by dots(.).

- A) 8-bit fields
- B) 4-bit fields
- C) 6-bit fields
- D) 2-bit fields

xiv) Why should network administrators subtract two addresses when determining the number valid hosts in each subnet?

- A) To cater for redundant default gateways (routers)
- B) To cater for the two addresses required for DHCP operation
- C) To cater for the subnet ID and default gateway (router)
- D) To cater for subnet broadcast address and subnet ID

xv) The rationale for subnetting is that

- A) It divides one large network into several smaller ones
- B) It divides network into network classes
- C) It speeds up the speed of network
- D) None of above

xvi) You spend your time as a network engineer thinking about the complete Class A network 10.16.0.0 and how you should subnet it. Finally, you decide how to subnet this Class A network, and you create a paper addressing and subnetting plan. Which of the following occurred to the parts of the structure of addresses in this new network?

- A) The subnet part got smaller
- B) The host part got smaller
- C) The network part got smaller
- D) The host part was removed
- E) The network part was removed

xvii) Which of the following is a true statement regarding the IP address 182.18.6.7?

- A) The network ID is 182.18.6.0
- B) The network is a Class A network
- C) The default mask for the network is 255.255.255.0
- D) The number of host bits in the unsubnetted network is 16

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viii) BBC Bank has been in business for the past 42 years. The bank provides a range of services including Cards that their customers can use to pay for services. Abdul Kisseka's credit card and information has been stolen by the Black Hat Hackers who call themselves WeHack. Abdul realised it and by the time he notified the bank, it was too late because someone had already used the card to shop for an item in the black market. Abdul has refused to come to terms that it is his fault. Which of the following security tenet will help protect BBC Bank in this case.

- A) Confidentiality
- B) Integrity
- C) Availability
- D) Non repudiation
- E) Authorisation

ix) In which class of IP address do we find a maximum of 254 Valid host addresses

- A) Class A
- B) Class B
- C) Class C
- D) Class D

x) You have been given 192.168.0.0 network address. Your task is to create two subnets (networks). How many bits do you require to create the two subnets?

- A) 1 Bit
- B) 2 Bits
- C) 3 Bits
- D) 4 Bits

xi) Which of the following is an example of a private IP address?

- A) 192.169.200.19
- B) 192.168.200.19
- C) 172.32.19.210
- D) 172.90.34.67

xii) You have been hired as a security consultant to review an organisation's security model. You find that the current security model establishes confidentiality such that system users can only read access control at their clearance level, the system also does not enable users with a certain level of clearance to view a document with a lower level of clearance and lastly the system does not prevent a user with a certain clearance level from writing data to a document rated higher than their clearance level. You are asked to design a new model to mitigate the status quo. Which IT security policy tenet will address this?

- A) Confidentiality
- B) Non-repudiation
- C) Integrity
- D) Availability

xxiv) Everything about computer networking can be summarised in TCP/IP suite or OSI model

A) TRUE: _____

B) FALSE: _____

xxv) Without routing and switching technologies, you would not be able to do this online real-time assessment

A) TRUE: _____

B) FALSE: _____

SECTION B: (ANSWER ANY THREE) [75 MARKS]

Question Two

Computer networks have changed the way information is sent from one device to another. In fact, without routing and switching, wireless and cable communications, we would not be able to communicate online.

a) What does the term computer network mean?

[01 Mark]

b) Offer any five examples that show the importance of data communications and computer networking in your daily life.

[10 Marks]

d) What distinguishes wireless networks from wired networks?

[04 Marks]

e) Does wired networks have advantages over wireless networks? Yes or No. Give reasons for your answer.

[10 Marks]

Question Three

The core of computer networking is understanding its components or the building blocks. Every aspect of networking can be linked to its components be it cables, wireless, servers, computers, http, BGP, services running on the network, users and so forth. In light of the above, explain any 5 key components of a computer network. Your answer should include the role the component plays, examples under such a component and where the component is placed on a network.

[25 Marks]

Question Four

a. Fill out the table below [10 marks]

Class	Range	Network bits	No. of networks	No. of hosts	Host bits
A			126	16777214	
B		16			16
C			2097150		

COMPUTER NETWORK FINALS

MAKERERE UNIVERSITY BUSINESS SCHOOL

FINAL EXAMINATION FOR THE DEGREE OF BACHELOR OF BUSINESS COMPUTING OF MAKERERE UNIVERSITY ACADEMIC YEAR 2022/2023

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: BUC1222

YEAR OF STUDY: ONE

SEMESTER: TWO

DATE: MONDAY, MAY 22, 2023,

TIME: 02.00 / 05.00 P.M

INSTRUCTIONS:

1. Section A is Compulsory
2. Answer all questions in section A
3. Answer any THREE questions in section B
4. All questions carry equal marks.
5. Follow any other instructions as provided during the examinations.

SECTION A: (COMPULSORY)

Question One (25 MARKS)

i) A wide area network refers to _

- | | |
|--|---|
| A) a single network | C) interconnection of local area networks |
| B) a vast collection of different networks | D) none of the mentioned |

ii) In the Local Area Network (LAN), there are two kinds of users, those whose devices connect to the LAN using a cable while others use wireless technology to connect (and no cable). Which of the following is true regarding the use of Ethernet in this LAN?

- A) Only the devices that use cables are using Ethernet
- B) Only the devices that use wireless are using Ethernet
- C) Both the devices using cables and those using wireless are using Ethernet
- D) None of the devices are using Ethernet

iii) Which of the following terms describe Ethernet addresses that can be used to send one frame that is delivered to multiple devices on the LAN?

- A) Burned-in address
- B) Unicast address
- C) Broadcast-multicast address
- D) Multicast address

iv) Application layer of a network _____

- A) consists of software being run on the computer connected to the network
- B) Establishes, maintain and terminates virtual circuits
- C) Defines user port into the network
- D) None of these

v) When data reaches the data link layer, the data link adds its own data on it. In network terms, such entity created when encapsulating data inside data-link layer headers and trailers is

- A) Data
- B) Chunk
- C) Segment
- D) Frame
- F) Packet

vi) Routing and route protocols are essential for data to reach its intended destination on any network be a LAN, WAN or MAN. A routing protocol hence

- A) Advertising unknown routes to neighbouring routers
- B) Learning routes for subnets directly connected to the router
- C) Forwarding IP packets based on a packet's destination IP address
- D) Learning routes and putting those routes into the routing table for routes advertised

vii) How does a network switch decide to forward a frame destined for a known unicast MAC address?

- A) It compares the unicast destination address to the bridging, or MAC address, table
- B) It compares the unicast source address to the bridging, or MAC address, table
- C) It forwards the frame out all interfaces in the same VLAN except for the incoming interface
- D) It compares the destination IP address to the destination MAC address
- E) It compares the frame's incoming interface to the source MAC entry in the MAC address table

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